

SHIVAM KUMAR YADAV

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SUMMARY

Computer Science graduate specializing in applied Generative AI and LLM-based systems. Built and evaluated multi-agent research frameworks, retrieval-augmented generation (RAG) pipelines, and document intelligence solutions with a focus on latency optimization, hallucination reduction, and real-world usability. Seeking entry-level AI/LLM engineering roles where rapid experimentation and production-minded thinking matter.

Projects

Autonomous Multi-Agent Research Framework ([GitHub Link](#))

Built a multi-agent research system to automate long-form research tasks that typically require manual web searches and synthesis, reducing researcher effort per task by ~60%.

- Engineered a state-driven orchestration layer coordinating Planner, Search, and Synthesizer agents using LangGraph's conditional routing to handle complex research workflows.
- Diagnosed sequential search as a critical bottleneck and implemented parallel execution, reducing end-to-end research latency by ~35% compared to baseline workflows.
- Validated framework reliability across 20+ diverse research queries spanning technical, legal, and market research domains, consistently producing structured reports with verifiable source citations.

Document Intelligence & RAG Pipeline ([GitHub Link](#))

Built an end-to-end RAG system to enable accurate Q&A over private document collections without requiring document-specific model fine-tuning.

- Engineered a complete document processing pipeline handling PDF ingestion, intelligent text chunking, vector embeddings generation, and semantic retrieval for contextual Q&A
- Systematically compared fixed-size chunking (500, 1000, 1500 tokens) against semantic splitting strategies, optimizing for context relevance while maintaining sub-5s query latency
- Reduced hallucination frequency by ~40% through targeted prompt engineering (system instructions + few-shot examples) and temperature tuning (0.3 vs 0.7 baseline)
- Validated the system on 15+ real-world document sets spanning legal contracts, technical manuals, and research papers, consistently achieving 2–5s response latency on consumer-grade hardware without GPU acceleration

Vision-Based Semantic Extraction & Translation([GitHub Link](#))

Developed an OCR application to automate extraction of structured data from medical prescriptions and pharmaceutical labels, reducing manual data entry time for pharmacy workflows.

- Built a vision-first extraction pipeline targeting high-variability medical documents where traditional template-based approaches fail.
- Optimized image preprocessing and visual prompting strategies through iterative testing, improving handwritten text recognition accuracy by ~15% over baseline Vision API configuration.
- Validated on a 100+ image test set with manual ground-truth annotation: achieved ~95% character-level accuracy on printed text and ~88% on handwritten samples.

TECHNICAL SKILLS

Generative AI & LLMs: LangChain, LangGraph, RAG Systems, Prompt Engineering, LLaMA-3, OpenAI GPT, Google Gemini, Multi-Agent Orchestration

NLP & Text Processing: SpaCy, NLTK, Semantic Search, Text Classification, Document Parsing

Data & Engineering: Python, SQL, Pandas, NumPy, ChromaDB (Vector DB), Hugging Face Transformers

Deployment & Tools: Streamlit, Gradio, GitHub Actions, n8n Workflows

EDUCATION

Bachelor of Technology (B.Tech)

Chouksey Engineering College

2025

XII (CBSE)

Burgess English Higher Secondary School

2021

CERTIFICATIONS & ACHIEVEMENTS

Data Science & Generative Professional – Oracle

AI Skills Passport – EY & Microsoft

AIML Data Analysis Workshop – ISRO (7-Day)

Machine Learning – IBM